DATASET METHODS REPORT

JULY 2026

Procedure Atlas: From HoloLens Recordings to Training-Ready Supervision

A general protocol for actions, execution mistakes, hand pose, and optional multi-modal evidence

Procedure Atlas / Multimodal Dataset Labeler
"Label the procedure. Ground the evidence." Human-in-the-loop preparation of egocentric recordings for reproducible procedural learning.

Scope	HoloLens and first-person recordings; temporal actions; structured mistakes; MediaPipe 21-landmark hands; optional object, tool, state, recovery, and sensor-reference evidence.
Priority model	P0 is always on. P1 is optional, reversible, and may be disabled when its evidence is unavailable.
Primary outputs	Project JSON, capture and temporal manifests, step and hand-keypoint CSV, COCO keypoints, split files, procedure JSON, and YAML.
Document status	Operational protocol and research-paper-style system report.

AIOps Project
Procedure Atlas schema 2.0 workflow

Abstract

Egocentric HoloLens recordings can support procedural action segmentation, execution-error recognition, and hand-aware learning only when raw media is converted into synchronized, auditable supervision. Procedure Atlas implements an always-on P0 contract comprising capture identity and split metadata, versioned action taxonomies, structured mistake semantics, presentation-timestamp-based intervals, reviewable MediaPipe hand keyframes, portable projects, and semantic validation. Optional P1 labeling adds overlapping object/tool interaction, contact, task state, recovery, task-graph, and HoloLens sensor-reference evidence. P1 is disabled by default and may be excluded from exports without deleting saved optional annotations. Screenshots use a 20-second proxy extracted from the supplied `20260716_012249_HoloLens.mp4`; their labels are illustrative, not adjudicated ground truth. This report defines the operating procedure, release gates, and remaining limitations for producing training-ready data.

Keywords: egocentric video; HoloLens; temporal action segmentation; procedural errors; hand pose; multimodal evidence; dataset curation

1 Objective and recommended identity

Recommended name: Procedure Atlas. This is broader and more accurate than "Hand Atlas" because the system labels a procedure, including its actions, outcomes, hands, objects, tools, and state evidence. The recommended header is **"Label the procedure. Ground the evidence."** It communicates that labels must be tied to observable or synchronized evidence, not inference.

Ego4D demonstrated the value of time-indexed wearable-camera data [3]; CaptainCook4D showed why procedural-error datasets need both normal and erroneous executions [4]. The target dataset should support: (1) action recognition and segmentation, (2) structured execution-error attribution, and (3) hand-aware learning. HoloLens Research Mode can expose depth, tracking-camera, pose, and inertial streams [1]. The current app stores P1 references to such streams but does not decode, calibrate, or synchronize them.

2 Implemented system

2.1 P0: always-on training contract

1. **Identity and provenance:** participant, session, split, nominal frame rate, time base, annotator, reviewer, taxonomy version, and guideline version.
2. **Temporal actions:** non-overlapping half-open intervals $[t_{\text{start}}, t_{\text{end}})$ with stable class IDs; remaining time exports as `background`.
3. **Structured outcomes:** incorrect segments require a mistake family, severity, expected and observed state, affected object, recovery action, and recoverability.
4. **Hand pose:** MediaPipe proposes 21 landmarks per hand [2]; reviewers correct position and visibility before saving a human-reviewed keyframe.
5. **Reproducibility:** autosave, undo/redo, portable save/open, editable segments, validation, and redundant exports.

Presentation time in seconds is canonical. A browser-presented-frame counter is separate; `source_frame_index` remains null unless an authoritative decoder or manifest supplies it. Exported `z_rel` is MediaPipe-relative depth, not calibrated metric HoloLens depth.

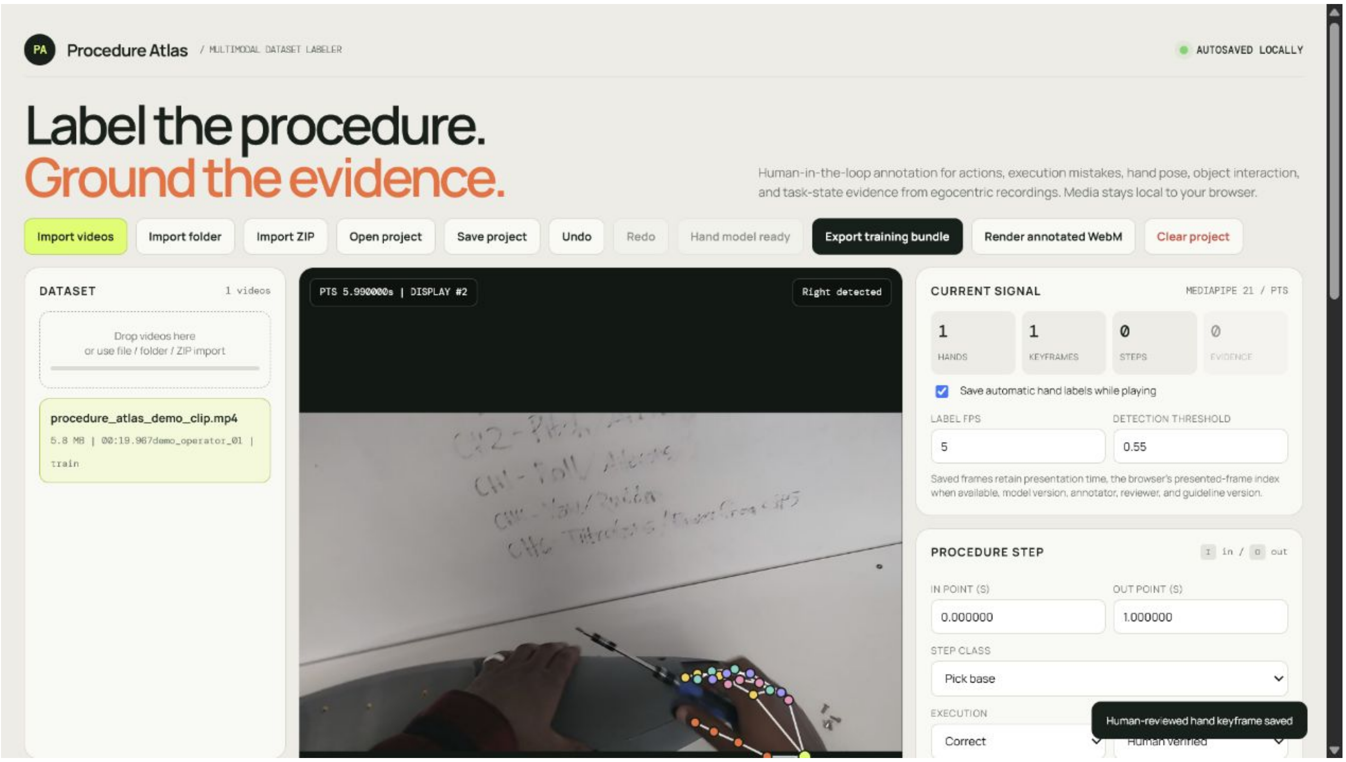
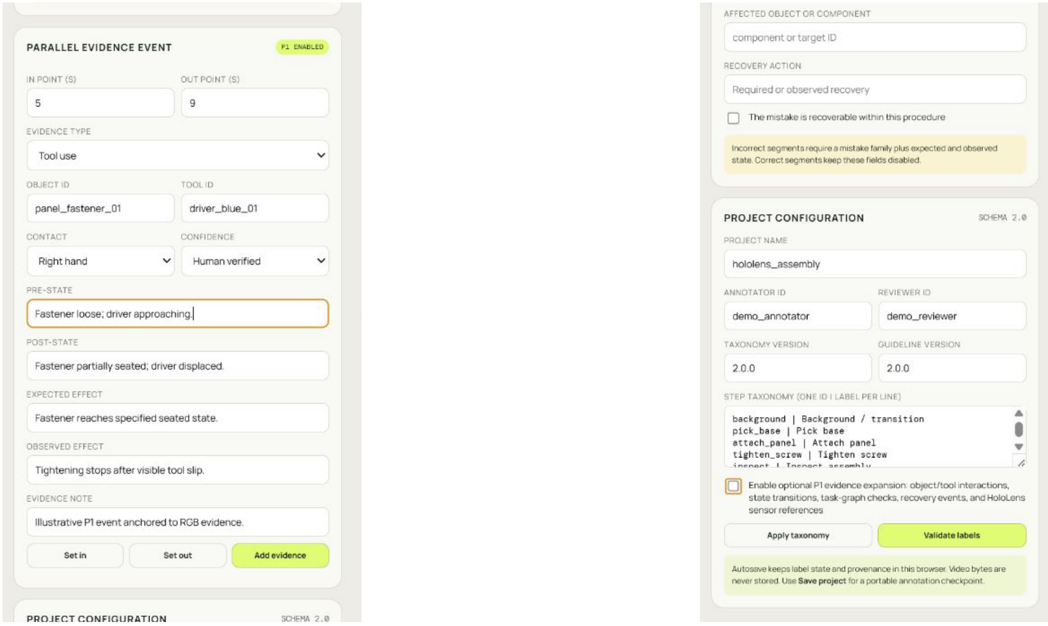


Figure 1. Procedure Atlas on the supplied HoloLens demo dataset. The real egocentric frame shows a driver interaction and a detected right-hand 21-joint overlay. The saved label is a workflow demonstration.

2.2 P1: optional evidence expansion

P1 adds overlapping events for object interaction, tool use, task-state transition, hand-object contact, recovery, and sensor anchors. Events can record object/tool IDs, contact hand, pre/post-state, expected/observed effect, confidence, and evidence notes. A task graph supports transition validation; optional fields reference depth, gaze, head pose/IMU, calibration, and timing.

P1 is **off by default**. Disabling it preserves its records in the project but excludes them from the training bundle. RGB-only datasets can therefore produce complete P0 supervision without presenting unsupported P1 fields as complete evidence.



(a) P1 enabled: overlapping tool-use evidence. **(b)** P1 disabled: the P0 project remains usable.

Figure 2. The optional expansion is controlled at project level and should be enabled only when its evidence can be labeled consistently.

3 Annotation and export contract

Table 1. Minimum decisions for each supervision channel.

Channel	Required content	Operational rule
Capture (P0)	participant, session, split, timing and versions	Use stable non-identifying IDs; split at participant/session level.
Action (P0)	class, start, end, status, confidence	Label the smallest complete unit; boundaries follow observable motion/effect.
Mistake (P0)	family, severity, expected/observed state, object, recovery	Use `incorrect` only when evidence contradicts the objective; otherwise use `unknown` or `aborted` as defined.
Hand (P0)	PTS, handedness, 21 points, visibility, source, reviewer	Inspect proposals and mark unlocalizable joints invisible rather than guessing.
Evidence (P1)	interval, type, object/tool/contact, state/effect	Permit overlap; disable P1 when coverage cannot support the target model.

Table 2. Schema 2.0 training-bundle artifacts.

Artifact	Purpose
`annotations/project.json`	Canonical project and preserved label state.
`metadata/capture_manifest.csv`	Capture identity, split, dimensions, duration, timing policy, and optional stream references.
`annotations/temporal_manifest.jsonl`	One completed temporal record per video.
`annotations/step_segments.csv`	Flat action and structured-mistake table.
`annotations/hand_keypoints.csv`	One row per joint with PTS and provenance.
`annotations/coco_hand_keypoints.json`	COCO-style keypoint records [5].
`splits/{train,val,test,exclude}.txt`	Reproducible split membership.
`configs/procedure.json`, `dataset.yaml`	Taxonomy, task graph, and training configuration.

4 Step-by-step labeling procedure

4.1 Prepare, import, and configure

Step 1. Inventory and preserve. Assign participant, session, task, take, device, and stream IDs. Retain immutable sources, checksums, time base, firmware, calibration, and stream inventory.

Step 2. Complete privacy review. Confirm consent, use, retention, access, and required face, screen, document, badge, or audio redaction [3, 8].

Step 3. Create an annotation proxy. Use a broadly decodable codec without replacing the original; retain proxy-to-source PTS mapping.

Step 4. Freeze the guideline. Publish class definitions, boundaries, examples, mistake rules, and review policy.

Step 5. Import or relink. Use *Import videos*, *Import folder*, or *Import ZIP*; confirm duration, decode, and content.

Step 6. Enter capture identity. Fill participant, session, split, nominal frame rate, and source time base.

Step 7. Enter provenance. Fill annotator/reviewer and taxonomy/guideline versions, apply the versioned taxonomy, and save a portable checkpoint.

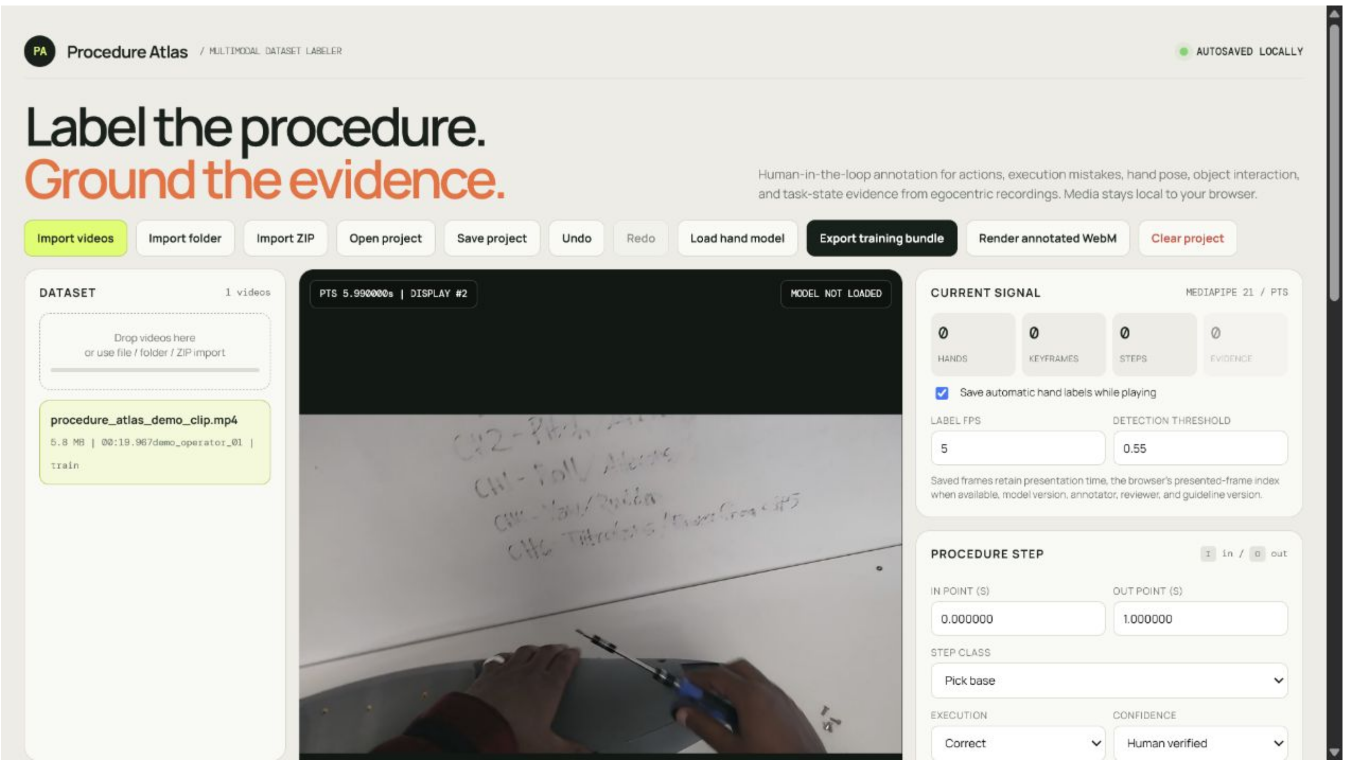


Figure 3. A 20-second proxy from `20260716_012249_HoloLens.mp4` loaded with P0 capture identity and train split assignment. Source footage remains local.

4.2 Label hands, actions, and mistakes

Step 8. Load the hand model. A failed model load is a processing failure, not evidence of absent hands.

Step 9. Generate and review hand proposals. Play with automatic capture enabled. Pause at contact, blur, occlusion, grasp change, or handedness uncertainty; correct joints and visibility and save the reviewed frame.

Step 10. Mark an action. Set *In* at the first task-directed motion and *Out* at the first frame after the observable effect. Select class, execution, and confidence, then add the segment.

Step 11. Ground a mistake. Record controlled family and severity, the expected state, observed contradictory evidence, affected object, and recovery. Do not infer hidden intent.

Step 12. Keep recovery separate. End the failed interval when its evidence ends; label later recovery as its own action and optional P1 event.

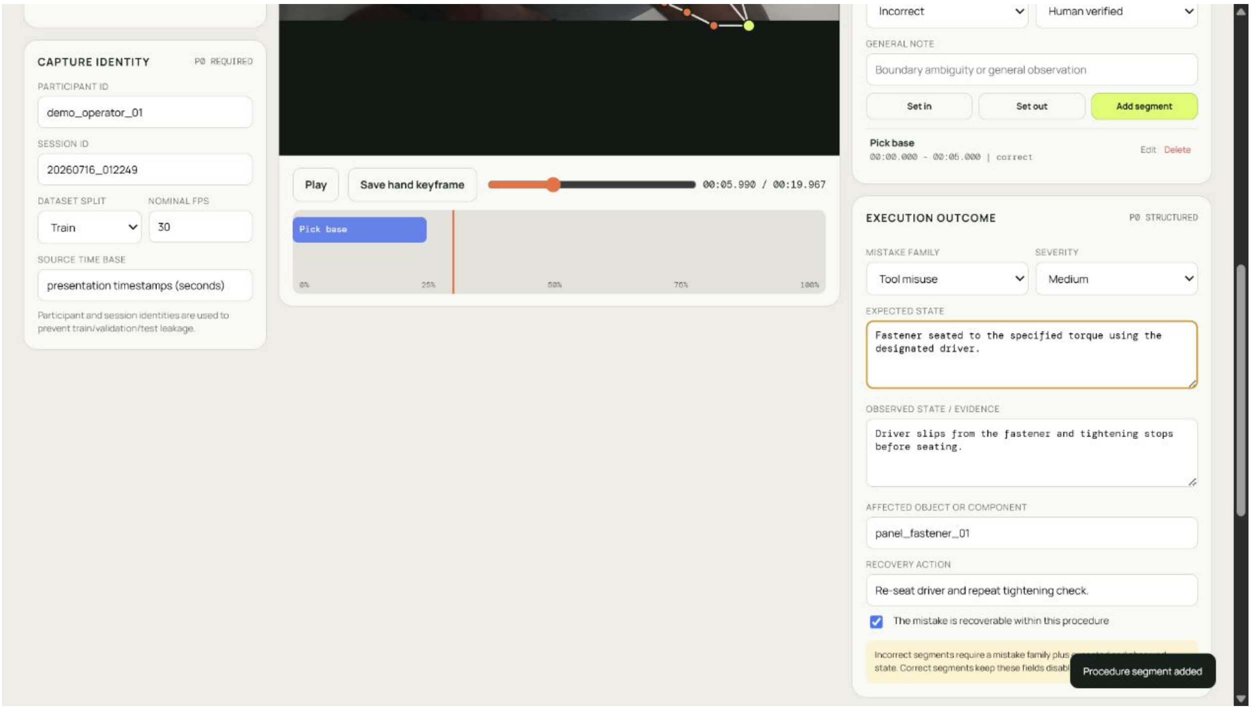


Figure 4. Structured P0 mistake entry for an illustrative *Tighten screw* segment. These fields are machine-readable. The example has not been adjudicated as ground truth for the recording.

4.3 Optional P1, validation, and export

Step 13. Decide whether P1 is supported. Enable it only when required object, state, recovery, or synchronized-stream evidence can be labeled consistently; otherwise leave it disabled.

Step 14. Add overlapping evidence. Record interval, type, object, tool, contact, pre-state, post-state, expected effect, observed effect, confidence, and note. Add optional stream references without claiming they were decoded.

Step 15. Validate. Resolve every error and disposition warnings for background, class coverage, optional sensors, and missing source indices.

Step 16. Review semantically. Schema validation does not prove boundary, mistake, hand, or state correctness. Independently review all test errors and a risk-weighted sample of other labels.

Step 17. Export and checksum. Record bundle, source and proxy, app and schema, taxonomy and guideline, annotator and reviewer, and review date.

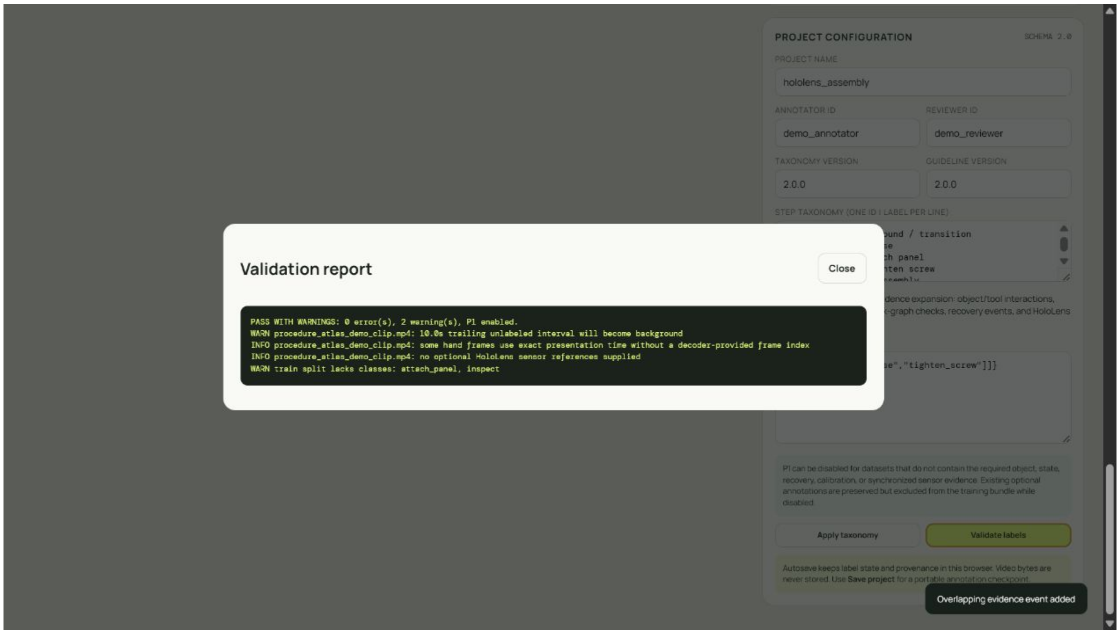


Figure 5. Validation reports zero errors while exposing actionable warnings and information: background duration, incomplete class coverage, missing decoder-provided frame indices, and absent optional HoloLens references.

5 Quality assurance and release gates

Independent review should oversample mistakes, rare steps, fast motion, occlusion, contact, and P1 events. Disagreements are adjudicated against the guideline and converted into new examples. Guidelines and inter-annotator agreement are coupled design problems [6].

Table 3. Recommended release gates; task risk may require stricter values.

Gate	Target	Meaning
Decode and identity	100%	Every source opens and has stable identity/timing.
Validation errors	0	Schema, timing, taxonomy, provenance, split, and graph pass.
Boundary agreement	median 0.25–0.50 s or better	Reviewers place comparable boundaries.
Mistake double-review	100% of test errors	Evaluation error claims are verified.
Hand audit	hard frames plus sampled baseline	Blur, contact, and visibility are reviewed.
Participant/session leakage	0	Evaluation represents unseen operators/sessions.
P1 completeness, if enabled	predefined target	Optional coverage supports the intended model.
Provenance	100%	Sources, labels, versions, and review history are linked.

6 Current limitations and required next work

P0/P1 schema 2.0 closes earlier gaps: controlled mistake fields, identity and review provenance, leakage checks, PTS/frame distinction, resumable projects, and parallel evidence. The following limitations remain material.

Table 4. Residual limitations affecting training readiness.

Priority	Limitation	Required response
P0	Source-to-proxy integrity is external.	Add source/proxy hashes, PTS mapping, capture-manifest import, and bundle checks.
P0	Semantic correctness/agreement are not automated.	Add adjudication state, disagreement workflow, boundary metrics, and review reports.
P0	Coverage warnings are coarse.	Add duration/count/mistake dashboards by participant, session, class, and split.
P0	Browser decoding lacks an authoritative source frame index.	Integrate a decoder/manifest mapping; keep the index null until authoritative.
P1	Sensor fields are references, not calibrated synchronized data.	Add stream ingestion, clock-drift estimation, calibration/pose validation, and synchronization quality metrics.
P1	Objects and contacts lack spatial geometry.	Add reviewed tracks, boxes/masks, stable object instances, and contact location.
P1	Hand depth is relative 2.5D, not metric 3D.	Fuse calibrated depth/pose and record coordinate frames and uncertainty.
P1	No central multi-user audit service exists.	Add access control, record locking, immutable revisions, and adjudication queues.

MediaPipe proposals also remain vulnerable to blur, gloves, unusual viewpoints, self-occlusion, and tool contact. P1 is therefore an expansion path, not a claim that all multimodal requirements are already supported.

7 Reproducibility and training-readiness checklist

```
dataset_root/
  raw/<participant>/<session>/<stream files>
  proxies/<participant>/<session>/<recording>.mp4
  calibration/<device>/<calibration version>/
  labels/<schema>/<taxonomy>/<export timestamp>/
  splits/{train,val,test,exclude}.txt
  metadata/capture_manifest.csv
  docs/{annotation_guideline,datasheet,consent_protocol}
```

The manifest should record source/proxy checksums, PTS mapping, identity, device, stream, frame rate/time base, calibration, consent/redaction, and exclusion reason. Publish a datasheet covering motivation, composition, collection, preprocessing, intended use, distribution, and maintenance [7]. Before training, verify that all errors are resolved; warnings are dispositioned; splits are participant/session-disjoint; test mistakes are adjudicated; unknown labels and invisible joints are masked; background is reviewed; COCO video/PTS references are materialized for image-only trainers; `z_rel` is not treated as metric depth; and P1 is disabled whenever its required evidence is incomplete.

8 Conclusion

Procedure Atlas provides a coherent P0 workflow for turning HoloLens RGB recordings into auditable temporal actions, structured execution outcomes, and reviewed 21-joint hand poses. Optional P1 evidence can extend this baseline without weakening it. The interface and validator are necessary but not sufficient: training readiness still depends on immutable source/proxy mapping, semantic review, leakage-resistant splits, coverage analysis, privacy controls, and—for multimodal or 3D objectives—calibrated synchronization and spatial annotation.

References

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